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Facts, empirical evidence if one prefers, do not exist independently of theories. Instead, the preconceptions of the investigator strongly influence his search for empirical evidence. Of course, preconceptions vary among investigators, and some are often quite amorphous. Nevertheless, excepting Marxists, most economists tend to group themselves around at least two significantly different sets of preconceptions. One set is generally shared by neoclassicists, the other by institutionalists. Overlapping is quite frequent, but the fact that the American Economic Association and the Association for Evolutionary Economics are separate national organizations is evidence of this grouping. The significantly different preconceptions of and types of evidence sought by the two groups are important distinctions between them, and these differences will be analyzed here in light of recent works in epistemology and methodology.[1]

Preconceptions

The character and effects of three related preconceptions concern us: (1) the kind of model or theory constructed, either descriptive or pre-

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dictive, (2) the unit of analysis used, either the institution or the individual, and (3) the psychological perspective of the theorist, either behaviorist or subjectivist. These preconceptions are not air-tight compartments into which all theorists must be squeezed. They are merely useful constructs to aid understanding. And, even though this writer is an institutionalist, the purpose of the following analysis is to understand the methodological differences between institutional and neoclassical economics, not to make invidious comparisons.

Pattern and Predictive Theory

Institutionalists seek to construct pattern models (theories), while neoclassicists seek to construct predictive models (theories).[2] A pattern model explains human behavior by carefully placing it in its institutional and cultural context. A predictive model explains human behavior by carefully stating assumptions and deducing implications (predictions) from them.

In neoclassical economics, a prediction is a logical deduction from a fundamental postulate or assumption. Hence, predictive evidence has to do with the empirical validity or accuracy of such deductions. As such, the nature of predictive evidence is easy enough to understand and requires little elaboration. However, a prediction is not necessarily a forecast. The former is logically deduced from an "assumption," while the latter is usually statistically derived from a model whose structure attempts to capture actual economic relations. Also, a forecast always refers to something in the future, but a prediction often refers to something in the past.

The distinction drawn here between different kinds of models is very similar to that drawn by Abraham Kaplan between concatenated (pattern) and hierarchical (predictive) theories. The components of a concatenated theory form an identifiable pattern (a culture), which generally converges to some central point (an institution). "A law or a fact is explained by a concatenated theory when its place is made manifest."[3] Individual behavior is explained when it is shown to fit into an institutional structure of behavioral norms, and institutional structure is explained when it is shown to fit into a cultural context. The components of a hierarchic theory are arranged in a pyramid, with basic principles or postulates (assumptions of utility and profit maximization) at the top and deductions drawn from them at lower levels in the theory. "A law is explained by the demonstration that it is a logical consequence of these principles, and a fact is explained when it is shown to follow from these along with certain initial conditions."[4] Individual behavior is explained when it is deduced from basic postulates (utility functions) and initial conditions (income distribution and prices of goods).

The predictive model is tested empirically by comparing deductions (quantitative predictions) with observations. The pattern model is tested empirically by comparing hypothesized institutional structures (qualitative patterns) with observations. In constructing pattern models, institutionalists spend great effort in making the pattern or theoretical structure realistic. In making predictive models, neoclassicists deemphasize structural realism. Instead, much effort is expended in testing predictions. According to Fritz Machlup, an important neoclassical practitioner, "the question is not whether firms of the real world will *really* maximize money profits, or whether they even *strive* to maximize their money profits, but rather whether the *assumption* that this is the objective of the theoretical firms in the artificial world of our construction will lead to conclusions—'inferred outcomes'—very different from those derived from admittedly more realistic assumptions."[5] The realism of these "inferred outcomes" or predictions is of major importance in neoclassical economics; not so in institutional economics, where the realism of a theory's structure or pattern is more important.

In summary, in the predictive mode, a theory is a set of predictions deduced or inferred from higher level principles or assumptions; in the pattern mode, a theory is a set of patterns which fit together. On the one hand, individual behavior is deduced from utility and income assumptions; on the other, individual behavior fits into an institutional structure, and institutional structure fits into a cultural context.

Institutions and Individuals as Units of Analysis

In generating their predictions, neoclassicists use the maximizing individual consumer or firm as their unit of analysis. In constructing their patterns, institutionalists use the institution.

To John R. Commons, one of the founders of institutional economics, an institution is a going concern which engages in a series of transactions within the guidelines of a set of working rules. An individual is both a part and a product of these going concerns. Individuals, pursuing their own interests *as individuals,* count for very little in the modern scheme of life. This is not to say the individual is of no causal significance. But it is to say that individuals realize their potential (or fail to do so) in going concerns, because it is through collective action that an environment is provided for effective individual action.[6]

To Thorstein Veblen, the other founder of institutionalism, an institution is a set of norms and ideals which is imperfectly (hence, subject to dramatic change) reproduced or internalized through habituation in each succeeding generation of individuals. Such an institution serves as stimulus *and* guide to individual behavior.[7] The preferences of an individual are not original or fundamental causal factors, hence they are not the place to start a theory. Rather, the nature of the world, in Veblen's view, is given by the sociological maxim that men do not simply do what they like, but they also like what they have to do.[8] Therefore, the place to start a theory (a pattern) is with what men have to do. Since, in the institutionalist view, the range of alternatives open to men is determined by the institutional structure or context into which they are born, the place to start is with that institutional structure.

Institutionalists try to keep their unit of analysis as realistic as possible because they are greatly concerned about descriptive realism. That is, they continuously ask how well their hypothesized institutional structures fit real-world observations. Their models are very detailed. Neoclassicists try to keep their unit of analysis as realistic as possible, too. However, their concern is with predictive realism. That is, they continuously ask how empirically accurate are their predictions. For example, in demand for money theory, much effort has been spent on comparing the empirical accuracy of the predictions derived from alternative demand functions.[9] The models are neat and clean—stripped to the bare logical bone. Occam's razor, in neoclassical hands, is used to cut away all the messy details (which institutionalists crave and neoclassicists abhor).

Neither the neoclassical firm nor the neoclassical consumer is a unit of analysis based on actual consumers or actual firms. Nor are they intended to be. Instead, each is a kind of "causal" link. According to Machlup, "in this causal connection the firm is only a theoretical link, a mental construct helping to explain how one gets from the cause to the effect."[10] Such a theoretical causal connection is used to generate predictions, not to build patterns. And the neoclassical consumer and firm, void of all the messy institutionalist realism, can better serve their *predictive* function.

In the case of institutionalists, both the firm and the consumer are based on actual corporations and consumers. The going concerns of Commons were based on his experiences and case studies. The leisure class described by Veblen was based on his acute, if acerbic, observations of our pecuniary culture. Such realistic units of analysis do not serve a *predictive* function, nor are they intended to. Instead, they help the economist understand the *pattern* of capitalistic American culture.

Perhaps the distinction between the institutional and neoclassical units of analysis will be clarified if the reader notes that the pattern models of institutionalists have much in common with anthropology and its unit of analysis (cultural pattern).[11] In contrast, the deductive models of neoclassicists have much in common with pure mathematics and its unit of analysis (logical postulate).

Behaviorism and Individualism as Psychological Perspectives

The institutional approach takes the psychological perspective of behaviorism.[12] Its significance for institutional economics is illustrated by the following passage from B. F. Skinner about the effect of institutional norms on the personality or subjective preferences of an individual scientist:

> If he designs an experiment in a particular way, or stops an experiment at a particular point, because the result will then confirm a theory bearing his name . . . he will almost certainly run into trouble [because] the published results of scientists are subject to rapid check by others. . . . To say that scientists are therefore more moral or ethical than other people, or that they have a more finely developed ethical sense [or to say that their individual preferences are skewed toward truth], is to make the mistake of attributing to the scientist what is actually a feature of the environment in which he works.[13]

Behaviorism grounds the roots of human action in institutional structures (norms, working rules, use and wont) rather than in individual preferences, which are considered to be either largely derivative or unreliable due to their introspective or subjective nature. Behaviorism is not void of preconceptions, despite Machlup's statement that "the program of behaviorism is to reject preconceptions and assumptions and rely only on observation of overt behavior."[14] Behaviorism preconceives individual preferences, when they must be used in an analysis, as largely derived from the cultural-institutional milieu into which the individual is born. Hence, an explanation of human behavior should start there.

However, this milieu and the associated individual behavior comprise more than a static set of observed stimuli and responses. The psychological preconceptions of institutionalists are far more sophisticated.[15] Individual human beings are also important, particularly when they struggle against and change the institutional boundaries within which they are born. Veblen stated, in an important article, that man as envisioned by orthodox economists "has neither antecedent nor consequent. He is an isolated definitive human datum, in stable equilibrium except for the buffets of the impinging forces that displace him in one direction or another. . . . When the force of the impact is spent, he comes to rest, a self-contained globule of desire as before."[16] Although institutionalists turn to behaviorism in their distrust of subjectivism, Veblen clearly envisioned man as more than simply the missing middle term in stimulus-response psychology.

If the investigator begins explaining human behavior with the individual and not the institution, he shares the subjective preconceptions of the neoclassicist, perhaps better known as methodological individualism. Paul Diesing defines this as "the doctrine that only individuals are real, societies and groups are not, and therefore all explanation must be based ultimately on statements or laws about individual behavior."[17] Diesing's definition is rather sharply drawn. Social scientists are rare indeed who argue that societies are not real. Rather, many (neoclassicists included) study societies by beginning with individuals and their independent choices.[18]

Aided by this discussion of preconceptions, institutional economics can be understood as a set of concatenated theories or pattern models composed of institutions as the building blocks and with behaviorism as the psychological foundation. In contrast, neoclassical economics can be understood as a set of hierarchic theories or predictive models composed of individual firms and individual consumers as the building blocks, with subjectivism or methodological individualism as the psychological foundation.[19]

Corresponding Types of Evidence

Not only are the preconceptions of the two schools quite different, but also institutional and neoclassical economists seek different kinds of evidence, namely, structural and predictive. Structural evidence has to do with how well the hypothesized pattern model or concatenated theory of the institutionalist fits the set of human relations he is trying to explain. Predictive evidence has to do with the accuracy and reliability of predictions derived from the hierarchic theory or predictive model of the neoclassicist.

A simple example will illustrate.[20] Suppose one wishes to investigate the strength of a chair, but cannot sit on it. Two kinds of evidence could be gathered. The chair's structure could be examined by observing the number, thickness, and strength of the legs, and the general state of repair. Or one could assume that the chair is sound, predict that people could use it safely, and then, in the course of observation, see how many people sit on it without the chair collapsing. In the first instance, structural evidence has been gathered; in the second, predictive evidence.

Structural Evidence and Institutional Economics

Prediction is relatively unimportant in institutional economics. According to Charles Wilber and Robert Harrison, "the accuracy of predictions cannot be the main form of verification in the pattern model."[21] In such a model the behavior of individuals is not precisely determined by the pattern (institutional structure). Instead, a range is set within which the constituent parts can vary. In particular, institutions determine a spectrum of acceptable alternatives from which individuals can choose, but they do not predetermine exact behavior. Hence, precisely deduced predictions are often not possible when using an institution as the basic unit of analysis.

To give a trivial, but highly instructive, example: Knowing the norms of dress established in a particular organization does not give the theorist the power to predict *exactly* what Mr. Jones will wear on Monday. Nevertheless, an institutionalist would argue that although his coat might be either dark gray or pale brown, if Jones is an IBM executive, he *will* wear a suit and tie. Also, he will live in a particular kind of house and neighborhood, even though his exact address is problematic. Jones will drink scotch, not beer; he will marry a thin woman, not a fat one; and he will belong to the country club, not the bowling league. In short, an understanding of the institutional structure in which Jones is embedded does give the institutionalist some power to make *general,* qualitative predictions, but not *specific,* quantitative ones.

Institutionalists, naturally enough, generally seek evidence regarding institutional norms of behavior, rewards and sanctions, and power relations. All these have to do with institutional structure and cultural context. Wilber and Harrison sum up the institutionalist's search for validation: "Evidence in support of an hypothesis or interpretation is evaluated by means of contextual validation. This technique is a process of cross-checking different kinds of sources of evidence [historical studies, questionnaires, case studies, and so forth], and it serves as an indirect means of evaluating the plausibility of one's initial interpretations."[22] An institutionalist's theory of consumption uses institutional norms and cultural context to construct a pattern of consumer behavior. To test the theory empirically, an institutionalist uses case studies, historical studies, and/or questionnaires (if available) to see if the norms and context of the pattern model constructed coincide with the structure of the "real" world as observed and interpreted by himself and by others.

In institutionalism, what Thomas Kuhn would call "normal" science consists of conducting case studies and using them to elaborate on or extend a basic pattern.[23] "Revolutionary" science consists in constructing new basic patterns to replace old ones which no longer can be made to fit the "real" world.

For example, in his *Theory of the Leisure Class* and *Theory of Business Enterprise,* Veblen constructed a basic pattern model of pecuniary-predatory canons of right conduct in consuming and producing under modern capitalism. His *Imperial Germany* and *Higher Learning in America,* among others, were case studies in which he elaborated on and extended his pattern model to test how well it fit the world around him. In these case studies, he attempted to gather and interpret structural evidence. He made no concrete predictions and sought no predictive evidence. Rather, through his keen observations and voracious reading in all the social sciences, he sought to understand, not predict.

Predictive Evidence and Neoclassical Economics

The pattern models of institutional theory do not facilitate the generation of logical deductions or predictions,[24] but the deductive models of neoclassical theory do. Understandably, then, the main thrust of neoclassical economics has been toward logically generating deductions and empirically seeking predictive evidence.[25]

"Normal" science in neoclassical economics consists of gathering price and quantity data to test demand and supply predictions generated by deductive models of the consumer and producer. The "normal" science of institutional economics involves case studies along the lines of, say, Adolf Berle and Gardiner Means,[26] whereas that of neoclassical economics seeks to generate logical deductions by mathematical economists and statistically test these predictions through the work of their fellow economists. Of course, the mathematical deduction and empirical testing often occur simultaneously. For example, two rather remarkable works in neoclassical economics were published in 1938: R. G. D. Allen's *Mathematical Analysis for Economists* and Henry Schultz's *The Theory and Measurement of Demand.* Allen developed the mathematical tools required for quantitative deductions (predictions) and taught them to a generation of neoclassicists. Schultz, at the same time, pioneered in the statistical determination and testing of demand curves.[27]

Conclusion

This discussion attempts to clarify the methodological differences between institutionalists and neoclassicists. Distinctions were drawn between the preconceptions and types of evidence sought by the two schools of thought. The first constructs pattern models of institutional norms embedded in a specific cultural context. Within this model, the behavior of individuals is largely a result of the institutional milieu into which they are born. Hence, the detailed nature or structure of that institutional-cultural environment is of paramount theoretical and empirical importance. Case studies and questionnaires which probe this structure are the grist of this school's empirical mill. The neoclassical school constructs deductive models based on postulates which generate logical predictions about individual consumers and firms. Within this model, the optimization of individual choices is used to predict quantities supplied and demanded. Hence, the institutional-cultural environment is much less significant. Instead, statistical price and quantity data with which to compare predictions become of paramount importance.

It should be apparent why the two schools of thought, or the two paradigms, are so different. Their preconceptions and empirical searches take the practitioner in very different directions.

Notes

1. The works in the philosophy of science from which this article draws include: Paul Diesing, *Patterns of Discovery in the Social Sciences* (New York: Aldine-Atherton, 1971); Abraham Kaplan, *The Conduct of Inquiry* (San Francisco: Chandler, 1964); Ernest Nagel, *The Structure of Science* (New York: Harcourt, Brace and World, 1961); Stephen C. Pepper, *World Hypotheses: A Study in Evidence* (Berkeley: University of California Press, 1966); and Charles K. Wilber with Robert S. Harrison, "The Methodological Basis of Institutional Economics: Pattern Model, Storytelling, and Holism," *Journal of Economic Issues* 12 (March 1978): 61–89. See also Thorstein Veblen, "The Preconceptions of Economic Science," in *The Place of Science in Modern Civilization and Other Essays* (New York: B. W. Huebsch, 1919).
2. This distinction draws heavily upon Diesing, *Patterns of Discovery;* Kaplan, *Conduct of Inquiry;* and Wilber and Harrison, "Methodological Basis."
3. Kaplan, *Conduct of Inquiry*, p. 298.

4. Ibid., p. 298. Also see pp. 332–46.
5. Fritz Machlup, "Theories of the Firm: Marginalist, Behavioral, Managerial," *American Economic Review* 57 (March 1967): 14–15. Machlup's position is very similar to the "F-twist." See Milton Friedman, "The Methodology of Positive Economics," in his *Essays in Positive Economics* (Chicago: University of Chicago Press, 1953).
6. John R. Commons, *The Economics of Collective Action* (New York: Macmillan, 1950), particularly pp. 117–19.
7. Thorstein Veblen, "The Limitations of Marginal Utility," in *The Place of Science in Modern Civilization,* pp. 242–43.
8. See Paul Streeten, "Appendix: Recent Controversies," in *The Political Element in the Development of Economic Theory,* edited by Gunnar Myrdal (New York: Simon and Schuster, 1954), p. 215.
9. See David E. W. Laidler, *The Demand for Money,* 2d ed. (New York: Dun-Donnelley, 1977), pp. 101–62.
10. Machlup, "Theories of the Firm," p. 9.
11. The interested reader should compare Ruth Benedict, *Patterns of Culture* (New York: Houghton Mifflin, 1934) with Thorstein Veblen, *Theory of the Leisure Class* (New York: Macmillan, 1899).
12. For a somewhat different but complementary view, compare with John E. Roemer, who states: "The classical economists and Marx placed *society* at the center of the economic system and the individual on the periphery. Neoclassical theory reverted to the Ptolemaic error by placing the individual at the center." John E. Roemer, "Neoclassicism, Marxism, and Collective Action," *Journal of Economic Issues* 12 (March 1978): 148.
13. B. F. Skinner, *Beyond Freedom and Dignity* (New York: Bantam Books, 1971), p. 166.
14. Machlup, "Theories of the Firm," p. 4.
15. This point was stressed by an anonymous referee who, although an institutionalist, is a gestaltist rather than behaviorist. For a discussion of introspection (subjectivism), behaviorism, and gestaltism, see Wolfgang Kohler, *Gestalt Psychology* (New York: New American Library, 1975 [1947]).
16. Thorstein Veblen, "Why Is Economics Not an Evolutionary Science?" in *Science in Modern Civilization,* pp. 73–74.
17. Diesing, *Patterns of Discovery,* p. 125. Also see Thorstein Veblen, "The Limitations of Marginal Utility," pp. 237–39 and "The Socialist Economics of Karl Marx and His Followers," pp. 441–43, both in *Science in Modern Civilization.* A discussion of Veblen's behaviorism and anti-subjectivism is found in Joseph Dorfman, *Thorstein Veblen and His America,* 7th ed. (Clifton, N.J.: Kelley, 1972), pp. 291–93.
18. See also Nagel, *Structure of Science,* pp. 535–46; and Kendall P. Cochran, "Economics as a Moral Science," *Review of Social Economy* 32 (October 1974): 185–95.
19. See also William M. Dugger, "Institutional and Neoclassical Economics Compared," *Social Science Quarterly* 58 (December 1977): 449–61.
20. The following example builds on the distinction and illustration drawn by Pepper, *World Hypotheses,* pp. 47–90.

21. Wilber and Harrison, "Methodological Basis," p. 78.
22. Ibid., p. 76. See also Diesing, *Patterns of Discovery,* pp. 147–48.
23. Thomas S. Kuhn, *The Structure of Scientific Revolutions,* 2d ed. (Chicago: University of Chicago Press, 1970).
24. Not all neoclassical economists seek and use predictive evidence to the same degree. For the two extremes compare Friedman, "Methodology of Positive Economics," with G. L. S. Shackle, *Time in Economics* (Amsterdam: North-Holland, 1958).
25. Viewed from a different angle, there may be two main thrusts or methodological positions within neoclassical economics. See Henry W. Briefs, *Three Views of Method in Economics* (Washington, D.C.: Georgetown University Press, 1960).
26. Adolf A. Berle and Gardiner C. Means, *The Modern Corporation and Private Property,* rev. ed. (New York: Harcourt, Brace and World, 1968).
27. R. G. D. Allen, *Mathematical Analysis for Economists* (London: Macmillan, 1938); and Henry Schultz, *The Theory and Measurement of Demand* (Chicago: University of Chicago Press, 1938).